

Supplementary information for

Facet sensitivity of iron carbides in Fischer-Tropsch synthesis

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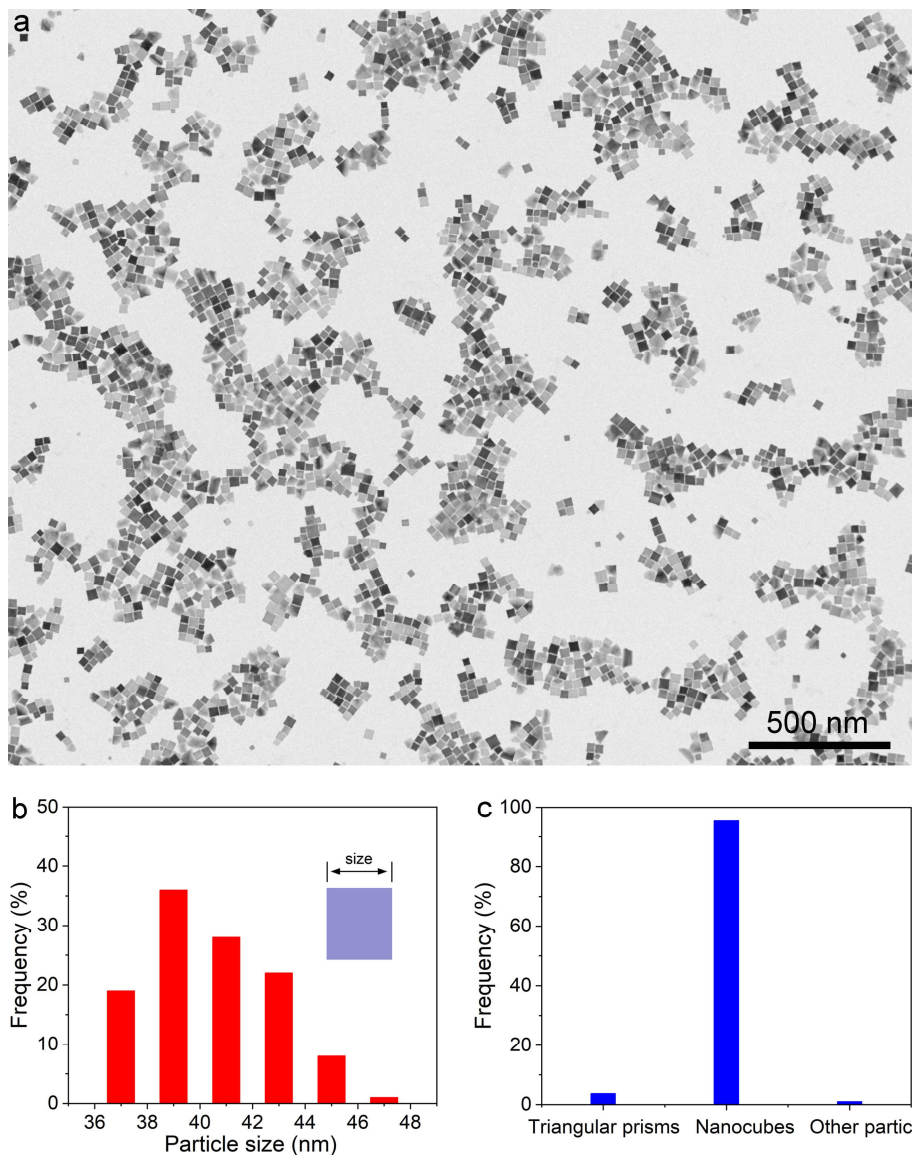
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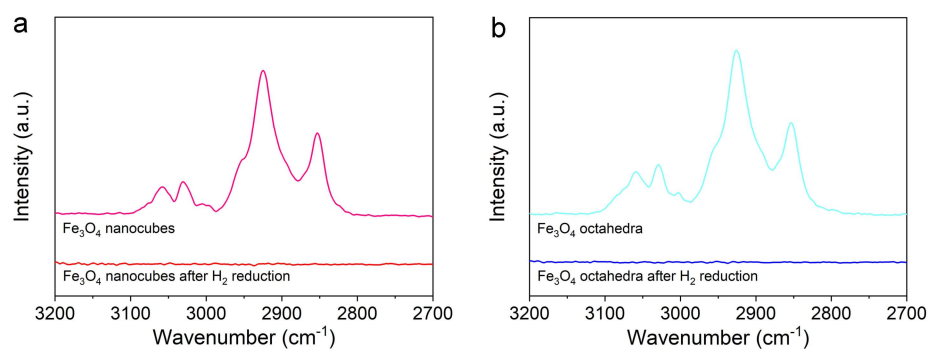
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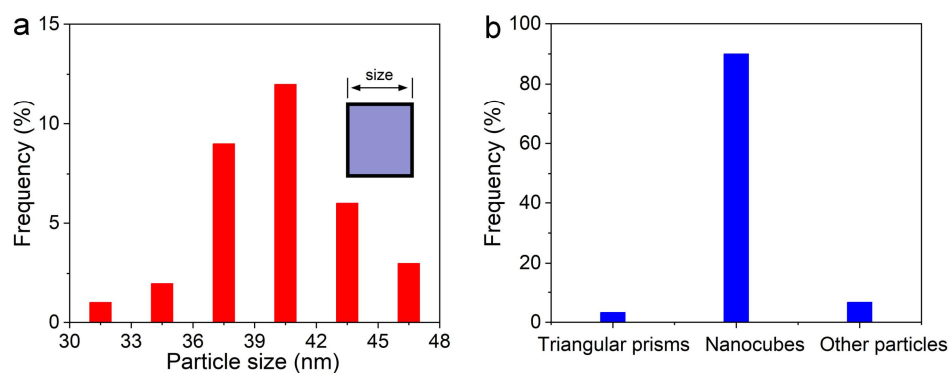
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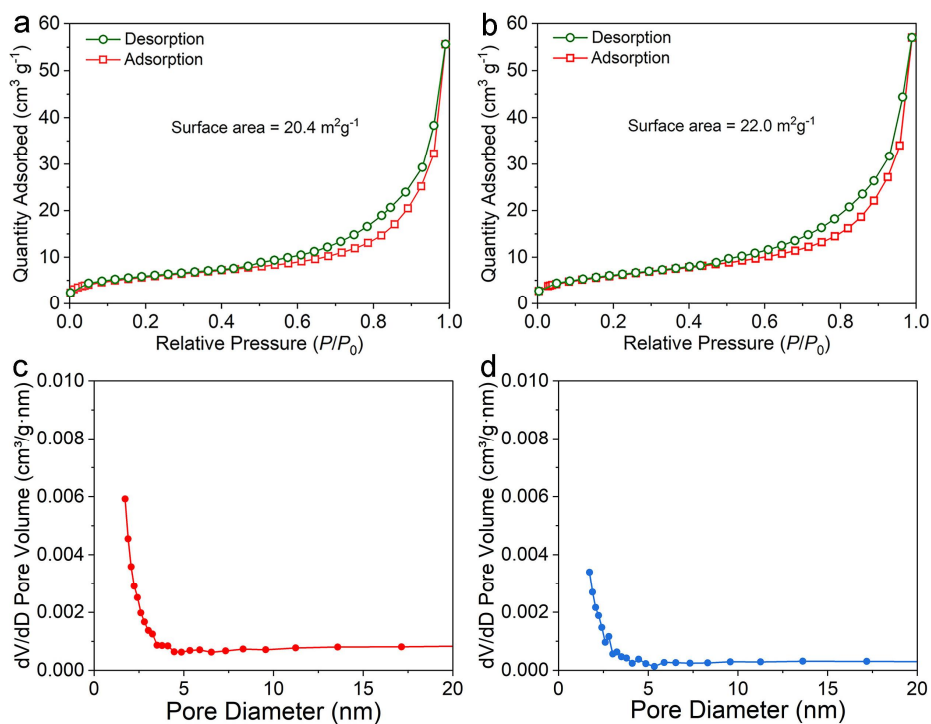
Supplementary Figure 1 | Structural characterizations of Fe₃O₄ nanocubes. (a) TEM image of Fe₃O₄ nanocubes. (b) Size distribution of the corresponding Fe₃O₄ nanocubes. The average particle size was 40.5±3.9 nm. (c) Contents of nanocrystals with different shapes. The purity of Fe₃O₄ nanocubes was 95.3%.



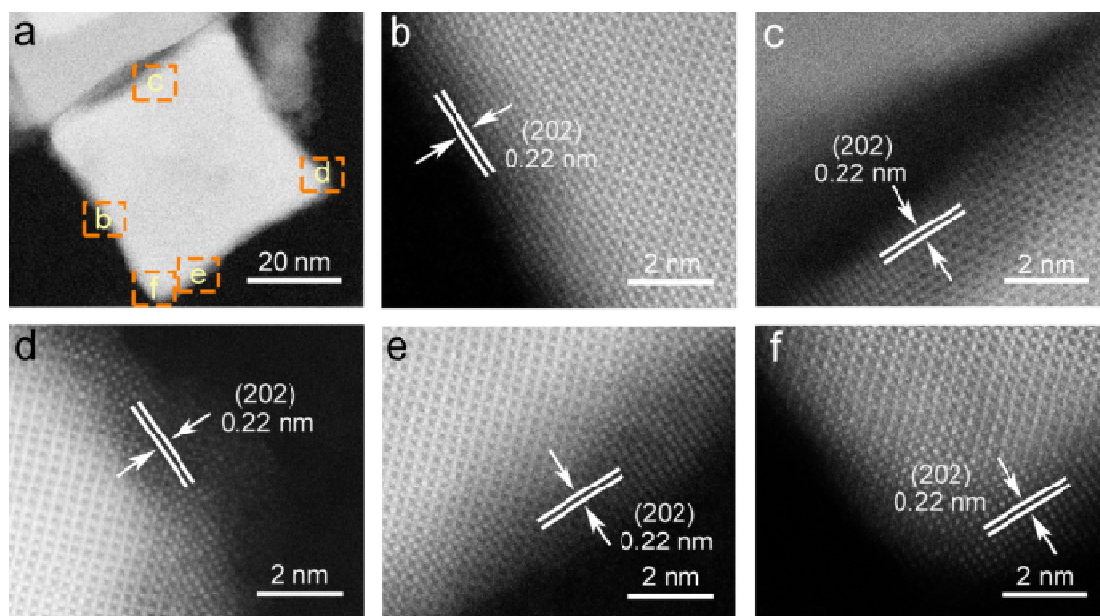
Supplementary Figure 2 | FTIR spectra. (a) FTIR spectra of Fe_3O_4 nanocubes before/after H_2 reduction. (b) FTIR spectra of Fe_3O_4 octahedra before/after H_2 reduction.



Supplementary Figure 3 | Statistics of $\text{Fe}_3\text{O}_4@ \chi\text{-Fe}_5\text{C}_2$ nanocubes from Figure 1a. (a) Size distribution of $\text{Fe}_3\text{O}_4@ \chi\text{-Fe}_5\text{C}_2$ nanocubes. The average particle size was 40.4 ± 3.8 nm. **(b)** Contents of nanocrystals with different shapes. The purity of $\text{Fe}_3\text{O}_4@ \chi\text{-Fe}_5\text{C}_2$ nanocubes was 90.0%.



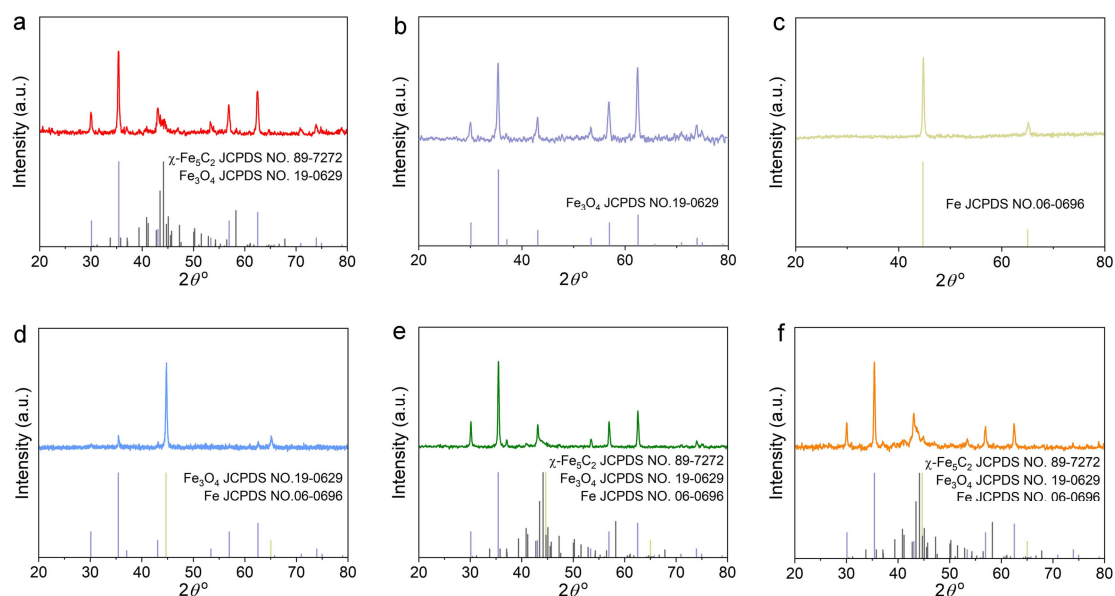
Supplementary Figure 4 | Textural properties of $\text{Fe}_3\text{O}_4@ \chi\text{-Fe}_5\text{C}_2$ nanocubes and octahedra before reaction. (a, b) Nitrogen adsorption and desorption isotherm of (a) $\text{Fe}_3\text{O}_4@ \chi\text{-Fe}_5\text{C}_2$ nanocubes and (b) $\text{Fe}_3\text{O}_4@ \chi\text{-Fe}_5\text{C}_2$ octahedra. (c, d) Pore-size distributions of (c) $\text{Fe}_3\text{O}_4@ \chi\text{-Fe}_5\text{C}_2$ nanocubes and (d) $\text{Fe}_3\text{O}_4@ \chi\text{-Fe}_5\text{C}_2$ octahedra derived from the nitrogen adsorption-desorption isotherms by the BJH method.



Supplementary Figure 5 | Structural characterizations of the $\text{Fe}_3\text{O}_4@\chi\text{-Fe}_5\text{C}_2$ nanocube. (a) HAADF-STEM image of an individual $\text{Fe}_3\text{O}_4@\chi\text{-Fe}_5\text{C}_2$ nanocube. (b-f) Magnified HAADF-STEM images of the region marked by the corresponding boxes in panel a.

Supplementary Table 1 | Mössbauer parameters of Fe₃O₄@ χ -Fe₅C₂ nanocubes and Fe₃O₄@ χ -Fe₅C₂ octahedra. The isomer shift (IS), quadrupole splitting (QS), hyperfine field, and spectral contribution are given.

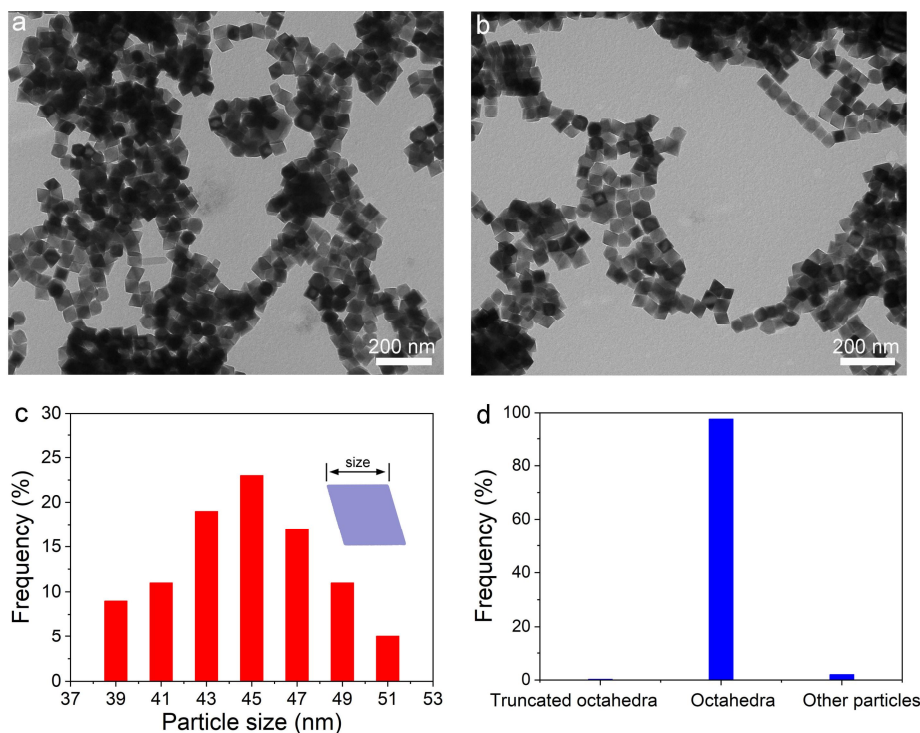
Samples	Phase ascription	Mössbaure parameters			
		IS (mm/s)	QS (mm/s)	Hyperfine field (T)	Spectral contribution (%)
Fe ₃ O ₄ @ χ -Fe ₅ C ₂ nanocubes	Fe ₃ O ₄ (A)	0.28	-0.01	48.8	47.5%
	Fe ₃ O ₄ (B)	0.65	0.01	45.7	15.9%
	χ -Fe ₅ C ₂ (A)	0.22	0.10	21.9	11.0%
	χ -Fe ₅ C ₂ (B)	0.20	-0.01	18.7	14.5%
	χ -Fe ₅ C ₂ (C)	0.19	-0.07	11.1	7.7%
	Fe(II)/Fe(III)	0.28	1.06	-	3.4%
Fe ₃ O ₄ @ χ -Fe ₅ C ₂ octahedra	Fe ₃ O ₄ (A)	0.32	-0.02	49.6	49.1%
	Fe ₃ O ₄ (B)	0.65	0.02	45.8	16.0%
	χ -Fe ₅ C ₂ (A)	0.24	0.10	21.7	13.5%
	χ -Fe ₅ C ₂ (B)	0.19	-0.02	18.1	10.8%
	χ -Fe ₅ C ₂ (C)	0.20	0.06	11.7	5.2%
	Fe(II)/Fe(III)	0.41	1.01	-	5.4%



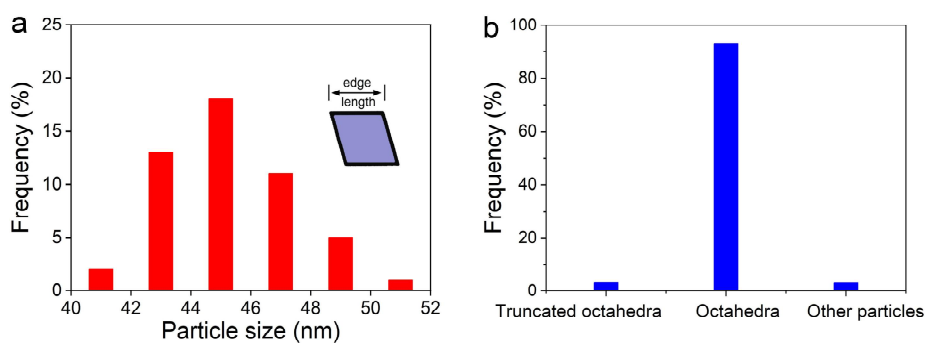
Supplementary Figure 6 | The corresponding XRD patterns of evolution from Fe₃O₄ nanocubes to Fe₃O₄@ χ -Fe₅C₂ nanocubes in Figure 2. (a) XRD pattern of Fe₃O₄@ χ -Fe₅C₂ nanocubes after the treatment of Fe nanocubes with syngas for 20 h. (b) XRD pattern of Fe₃O₄ nanocubes. (c) XRD pattern of Fe nanocubes. (d-f) XRD patterns of Fe₃O₄@ χ -Fe₅C₂ nanocubes after the treatment of Fe nanocubes with syngas for 2, 5, and 10 h, respectively.

Supplementary Table 2 | Quantitative analysis of the XRD patterns. The calculation was based on the Ratio of Intensity Reference (RIR) method.

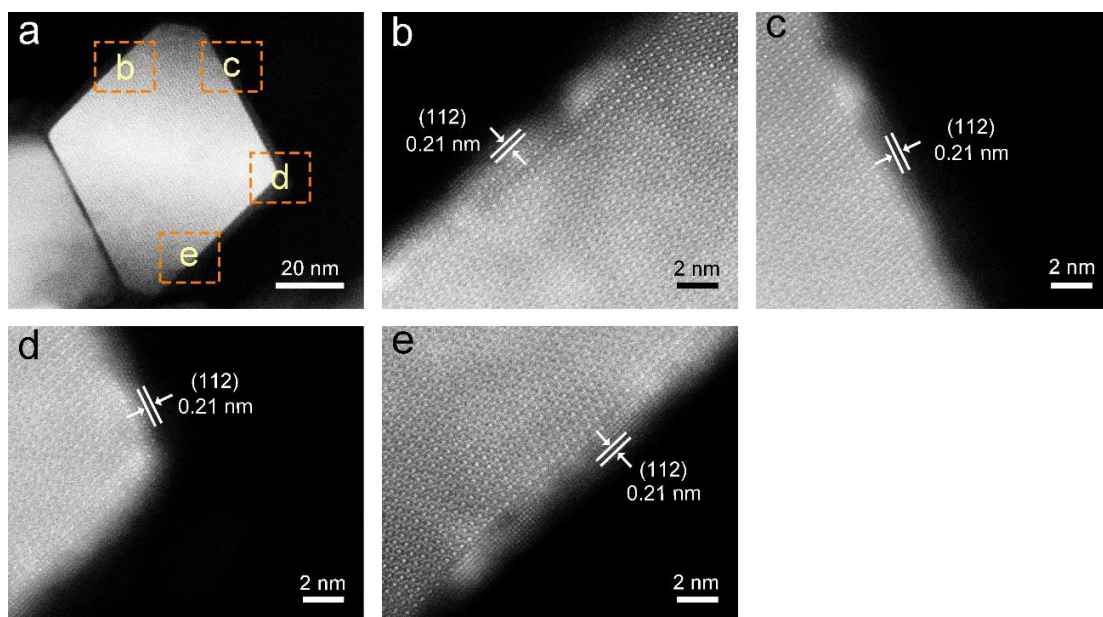
sample	Iron phase	Mass fraction (wt.%)
Fe ₃ O ₄ @ χ -Fe ₅ C ₂ nanocubes	Fe ₃ O ₄	70.2
	χ -Fe ₅ C ₂	29.8
Fe ₃ O ₄ @ χ -Fe ₅ C ₂ octahedra	Fe ₃ O ₄	72.4
	χ -Fe ₅ C ₂	27.6



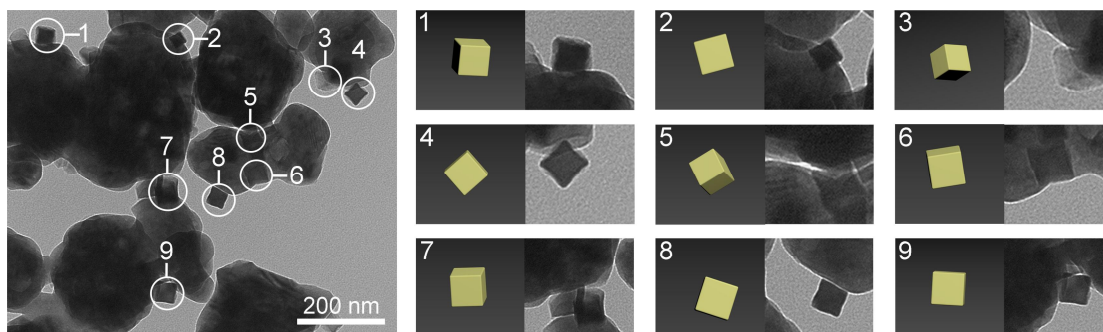
Supplementary Figure 7 | Structural characterizations of Fe₃O₄ octahedra. (a-b) TEM images of Fe₃O₄ octahedra. (c) Size distribution of the corresponding Fe₃O₄ octahedra. The average particle size was 44.8±3.3 nm. (d) Contents of nanocrystals with different shapes. The purity of Fe₃O₄ octahedra is 97.7%.



Supplementary Figure 8 | Statistics of $\text{Fe}_3\text{O}_4@ \chi\text{-Fe}_5\text{C}_2$ octahedra from Figure 3a. (a) Size distribution of $\text{Fe}_3\text{O}_4@ \chi\text{-Fe}_5\text{C}_2$ octahedra. The average particle size is 45.4 ± 3.5 nm. **(b)** Contents of nanocrystals with different shapes. The purity of $\text{Fe}_3\text{O}_4@ \chi\text{-Fe}_5\text{C}_2$ octahedra is 93.3%.

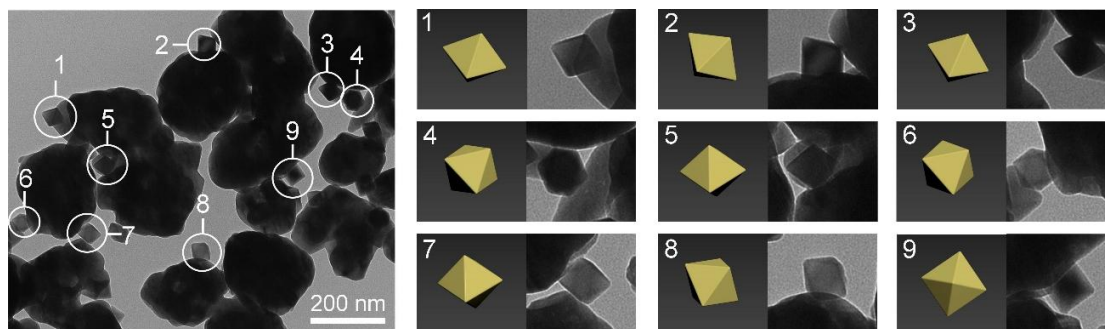


Supplementary Figure 9 | Structural characterizations of an individual $\text{Fe}_3\text{O}_4@\chi\text{-Fe}_5\text{C}_2$ octahedron/SiC. (a) HAADF-STEM image of an individual $\text{Fe}_3\text{O}_4@\chi\text{-Fe}_5\text{C}_2$ octahedron/SiC. (b-e) Magnified HAADF-STEM images of the region marked by the corresponding boxes in panel a.



Supplementary Figure 10 | Structural characterizations of $\text{Fe}_3\text{O}_4@ \chi\text{-Fe}_5\text{C}_2$ nanocubes/SiC.

TEM image of $\text{Fe}_3\text{O}_4@ \chi\text{-Fe}_5\text{C}_2$ nanocubes/SiC and the cubic models from different orientations.



Supplementary Figure 11 | Structural characterizations of $\text{Fe}_3\text{O}_4@\chi\text{-Fe}_5\text{C}_2$ octahedra/SiC. TEM image of $\text{Fe}_3\text{O}_4@\chi\text{-Fe}_5\text{C}_2$ octahedra/SiC and the octahedral models from different orientations.

Supplementary Table 3 | Catalytic properties of Fe₃O₄@ χ -Fe₅C₂ nanocubes/SiC and octahedra/SiC towards FTS.

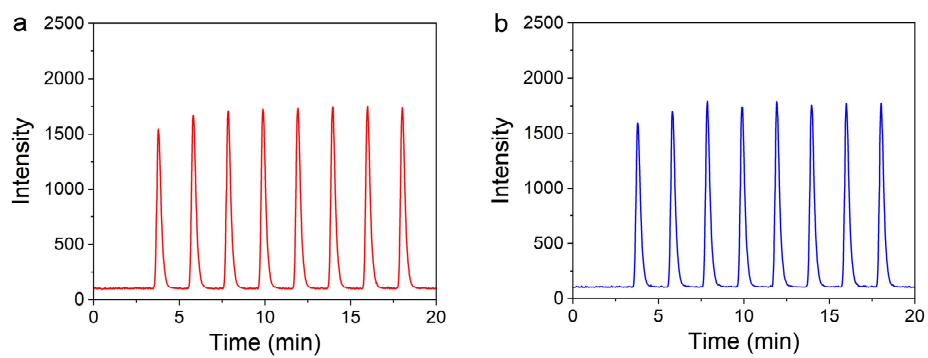
Catalysts	Conversion (%)	CO ₂ selectivity (%)	Product selectivity (C%, CO ₂ -free)						Carbon balance (%)
			CH ₄	C ₂ -C ₄ ⁼	C ₂ -C ₄ [°]	C ₅ -C ₁₂ ⁼	C ₅ -C ₁₂ [°]	C ₁₃ +	
Fe ₃ O ₄ @ χ -Fe ₅ C ₂ nanocubes/SiC ^a	45.4	18.7	14.2	21.6	19.4	17.9	24.3	2.6	98.7
Fe ₃ O ₄ @ χ -Fe ₅ C ₂ octahedra/SiC ^a	21.2	12.8	19.3	20.6	32.5	11.3	14.5	1.8	96.5
χ -Fe ₅ C ₂ nanoparticles/SiC ^a	23.6	10.5	15.5	29.8	18.4	16.9	18.0	1.4	96.8
Fe ₃ O ₄ @ χ -Fe ₅ C ₂ octahedra/SiC ^b	42.6	16.3	20.1	18.3	22.0	16.0	21.5	2.1	97.6

^a refers to the conditions of 20 bar, syngas (64 vol% H₂, 32 vol% CO, and 4 vol% Ar), 2400 mL h⁻¹ g_{cat}⁻¹, and 270 °C.

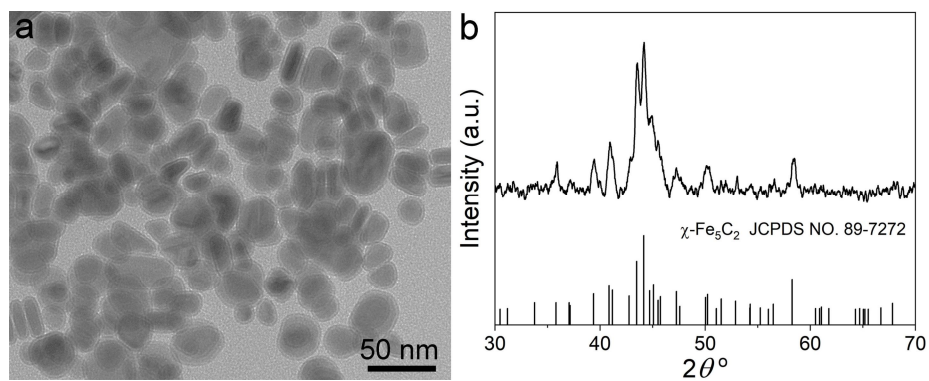
^b refers to the conditions of 20 bar, syngas (64 vol% H₂, 32 vol% CO, and 4 vol% Ar), 800 mL h⁻¹ g_{cat}⁻¹, and 270 °C.

⁼ refers to olefins.

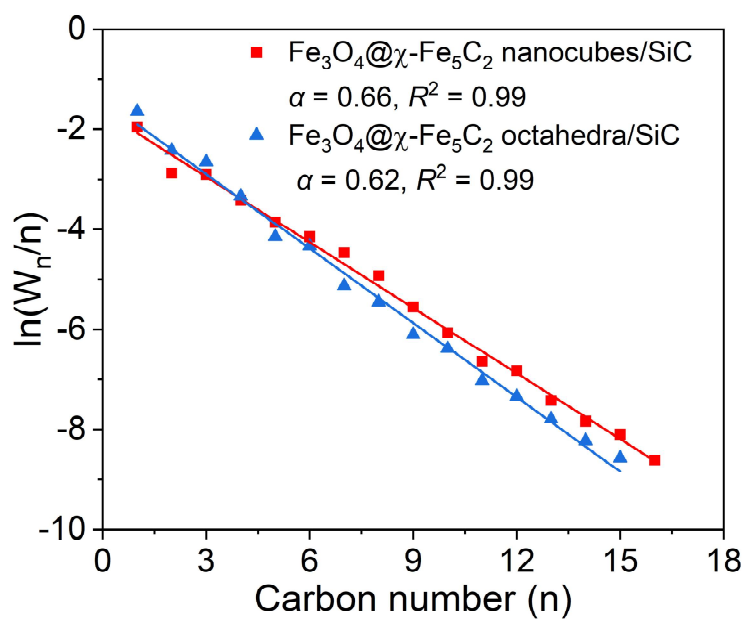
[°] refers to paraffins.



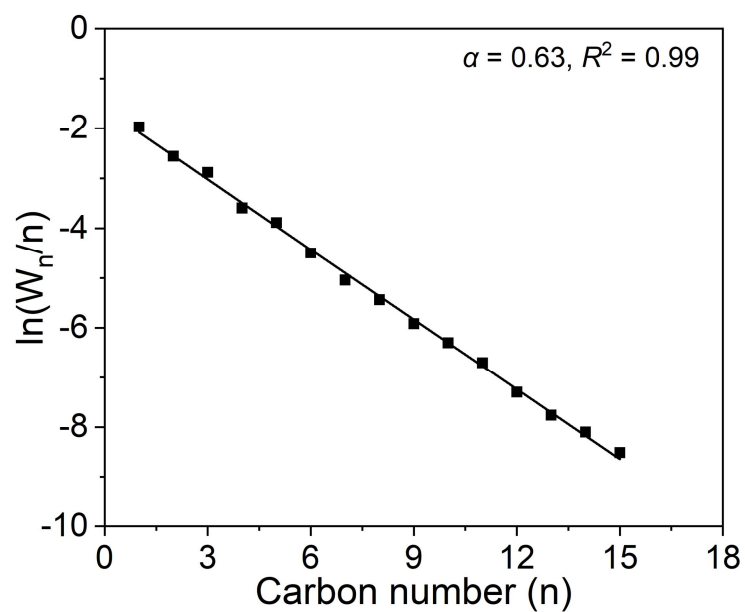
Supplementary Figure 12 | CO pulse profiles. CO pulse profiles of (a) $\text{Fe}_3\text{O}_4@\chi\text{-Fe}_5\text{C}_2$ nanocubes/SiC and (b) $\text{Fe}_3\text{O}_4@\chi\text{-Fe}_5\text{C}_2$ octahedra/SiC.



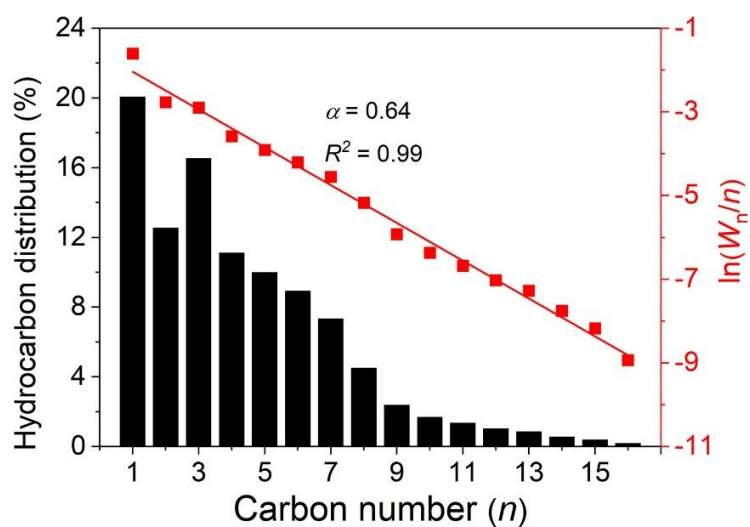
Supplementary Figure 13 | Structural characterizations of pure χ -Fe₅C₂ nanoparticles. (a) TEM image of pure χ -Fe₅C₂ nanoparticles. **(b)** XRD pattern of pure χ -Fe₅C₂ nanoparticles.



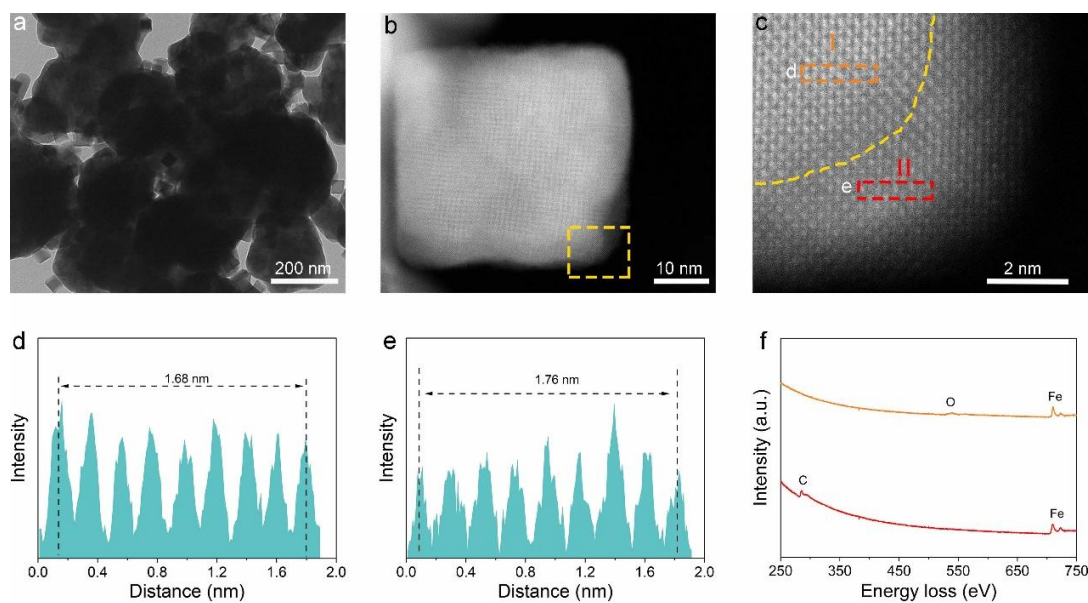
Supplementary Figure 14 | The ASF plot and the corresponding α value of $\text{Fe}_3\text{O}_4@ \chi\text{-Fe}_5\text{C}_2$ nanocubes/SiC and $\text{Fe}_3\text{O}_4@ \chi\text{-Fe}_5\text{C}_2$ octahedra/SiC. The reaction was conducted under 20 bar of syngas ($\text{CO}:\text{H}_2 = 1:2$, $2400 \text{ mL h}^{-1} \text{ g}_{\text{cat}}^{-1}$) at 270°C .



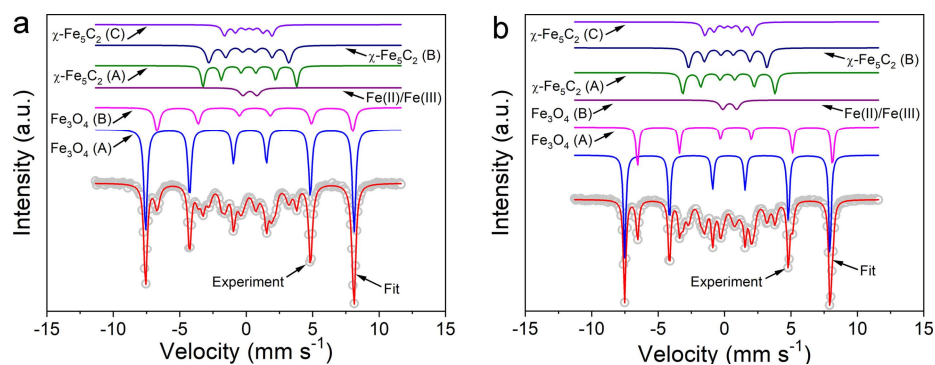
Supplementary Figure 15 | The ASF plot and the corresponding α value of χ -Fe₅C₂ nanoparticles/SiC. The reaction was conducted under 20 bar of syngas (CO:H₂ = 1:2, 2400 mL h⁻¹ g_{cat}⁻¹) at 270 °C.



Supplementary Figure 16 | Hydrocarbon distribution, the ASF plot, and the corresponding α value of $\text{Fe}_3\text{O}_4@\chi\text{-Fe}_5\text{C}_2$ octahedra/SiC at similar conversion levels of $\text{Fe}_3\text{O}_4@\chi\text{-Fe}_5\text{C}_2$ nanocubes/SiC. The reaction was conducted under 20 bar of syngas ($\text{CO}:\text{H}_2 = 1:2$, $800 \text{ mL h}^{-1} \text{ g}_{\text{cat}}^{-1}$) at 270°C .



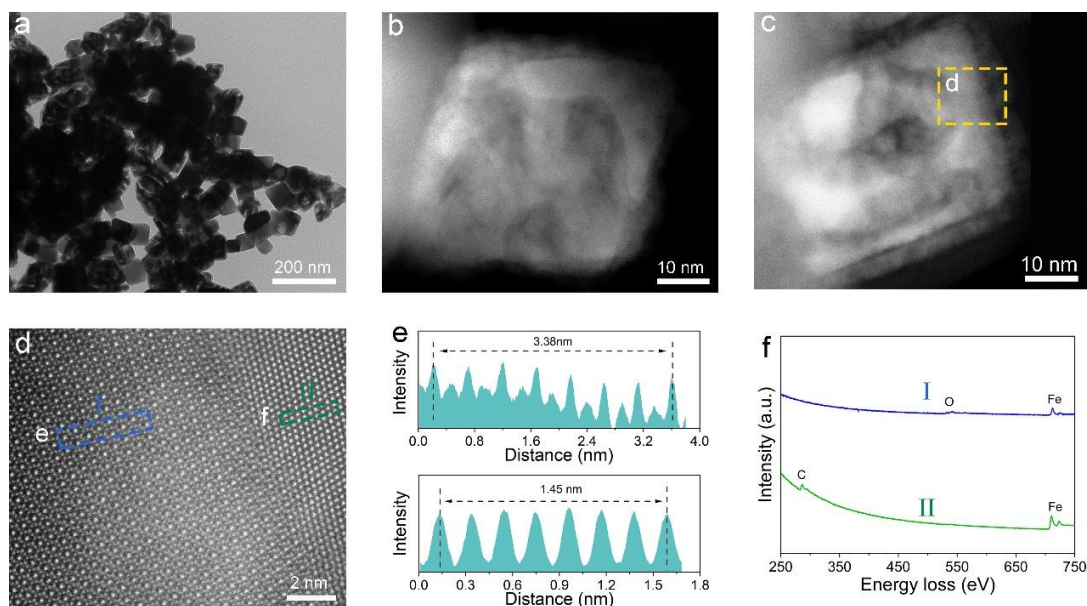
Supplementary Figure 17 | Structural characterizations of $\text{Fe}_3\text{O}_4@ \chi\text{-Fe}_5\text{C}_2$ nanocubes after 100 h on stream. (a) TEM image of $\text{Fe}_3\text{O}_4@ \chi\text{-Fe}_5\text{C}_2$ nanocube/SiC. (b) HAADF-STEM image of an individual $\text{Fe}_3\text{O}_4@ \chi\text{-Fe}_5\text{C}_2$ nanocube. (c) Magnified HAADF-STEM image of the region marked by the corresponding boxes in panel b. (d) Intensity profile recorded from the area indicated by the rectangular box in panel c. (e) Intensity profile recorded from the area indicated by the rectangular box in panel c. (f) EELS spectra of a $\text{Fe}_3\text{O}_4@ \chi\text{-Fe}_5\text{C}_2$ nanocube in panel c.



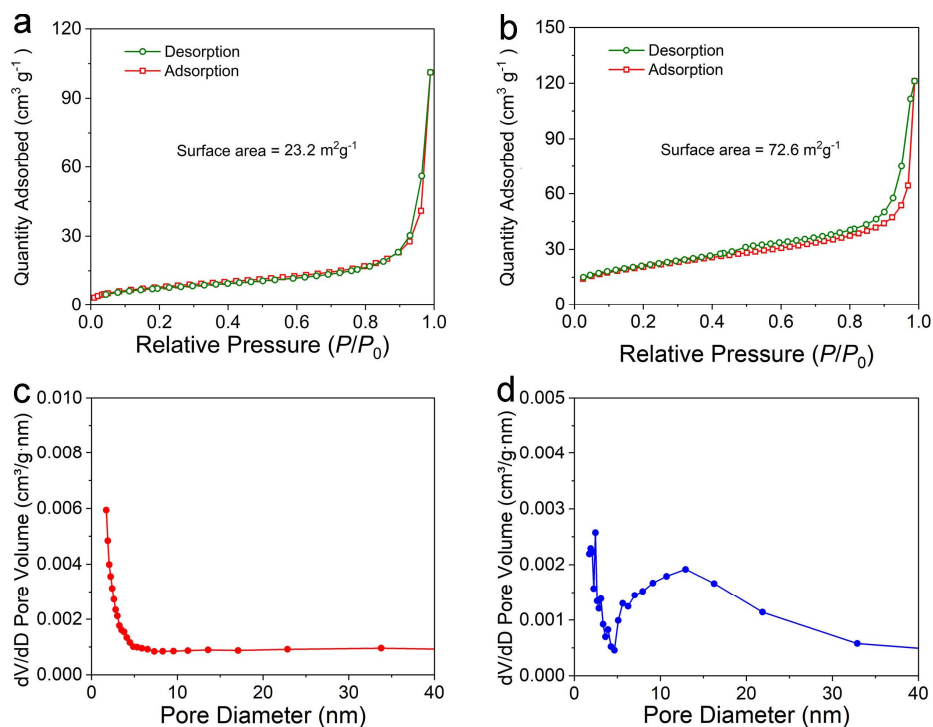
Supplementary Figure 18 | Mössbauer spectra characterizations. (a) Mössbauer spectra of $\text{Fe}_3\text{O}_4@\chi\text{-Fe}_5\text{C}_2$ nanocubes after 100 h on stream. **(b)** Mössbauer spectra of $\text{Fe}_3\text{O}_4@\chi\text{-Fe}_5\text{C}_2$ octahedra after 100 h on stream.

Supplementary Table 4 | Mössbauer parameters of Fe₃O₄@ χ -Fe₅C₂ nanocubes and Fe₃O₄@ χ -Fe₅C₂ octahedra after 100 h on stream. The isomer shift (IS), quadrupole splitting (QS), hyperfine field, and spectral contribution are given.

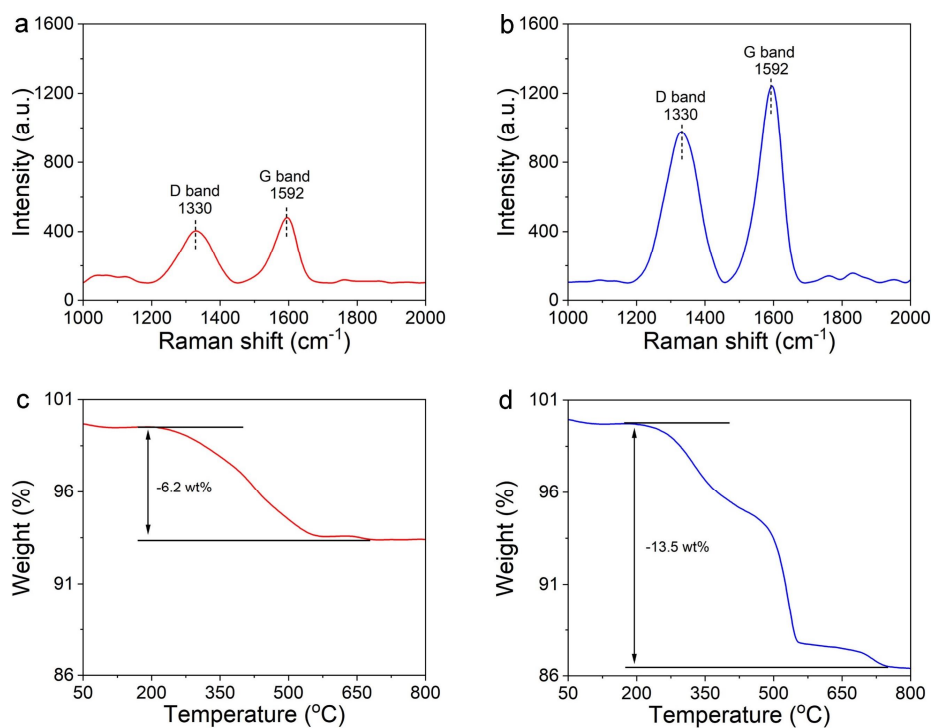
Samples	Phase ascription	Mössbaure parameters			
		IS (mm/s)	QS (mm/s)	Hyperfine field (T)	Spectral contribution
Fe ₃ O ₄ @ χ -Fe ₅ C ₂ nanocubes after reaction	Fe ₃ O ₄ (A)	0.30	-0.01	48.9	40.7%
	Fe ₃ O ₄ (B)	0.65	0.01	45.5	15.4%
	χ -Fe ₅ C ₂ (A)	0.22	0.11	21.8	12.9%
	χ -Fe ₅ C ₂ (B)	0.17	0.06	18.6	17.0%
	χ -Fe ₅ C ₂ (C)	0.20	0.02	11.1	9.9%
	Fe(II)/Fe(III)	0.34	1.07	-	4.1%
Fe ₃ O ₄ @ χ -Fe ₅ C ₂ octahedra after reaction	Fe ₃ O ₄ (A)	0.33	-0.05	48.5	38.7%
	Fe ₃ O ₄ (B)	0.78	-0.09	45.4	16.6%
	χ -Fe ₅ C ₂ (A)	0.27	0.11	21.6	14.5%
	χ -Fe ₅ C ₂ (B)	0.22	0.04	18.3	15.5%
	χ -Fe ₅ C ₂ (C)	0.28	0.06	11.2	10.4%
	Fe(II)/Fe(III)	0.39	1.04	-	4.3%



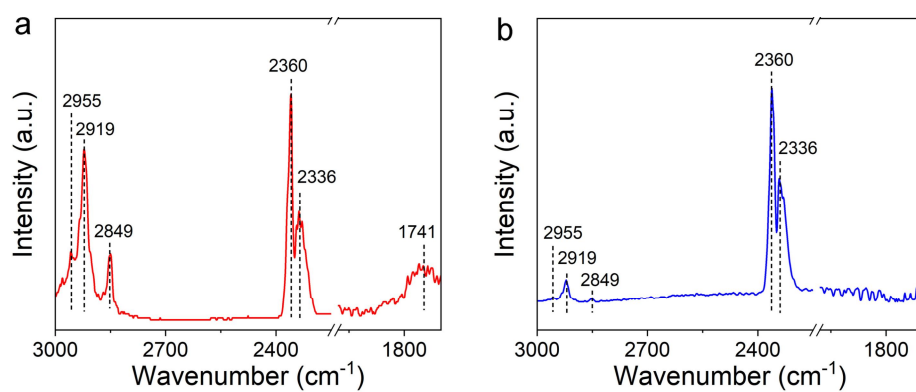
Supplementary Figure 19 | Structural characterizations of $\text{Fe}_3\text{O}_4@\chi\text{-Fe}_5\text{C}_2$ octahedra after 100 h on stream. (a) TEM image of $\text{Fe}_3\text{O}_4@\chi\text{-Fe}_5\text{C}_2$ octahedra/SiC. (b) HAADF-STEM image of an individual $\text{Fe}_3\text{O}_4@\chi\text{-Fe}_5\text{C}_2$ octahedron. (c) HAADF-STEM image of another $\text{Fe}_3\text{O}_4@\chi\text{-Fe}_5\text{C}_2$ octahedron. (c) Magnified HAADF-STEM image of the region marked by the corresponding boxes in panel b. (d) Intensity profile recorded from the area indicated by the rectangular box in panel d. (e) Intensity profile recorded from the area indicated by the rectangular box in panel d. (f) EELS spectra of a $\text{Fe}_3\text{O}_4@\chi\text{-Fe}_5\text{C}_2$ octahedron in panel d.



Supplementary Figure 20 | Textural properties of $\text{Fe}_3\text{O}_4@ \chi\text{-Fe}_5\text{C}_2$ nanocubes and octahedra after 100 h on stream. (a, b) Nitrogen adsorption and desorption isotherm of (a) $\text{Fe}_3\text{O}_4@ \chi\text{-Fe}_5\text{C}_2$ nanocubes and (b) $\text{Fe}_3\text{O}_4@ \chi\text{-Fe}_5\text{C}_2$ octahedra after 100 h on stream. (c, d) Pore-size distributions of (c) $\text{Fe}_3\text{O}_4@ \chi\text{-Fe}_5\text{C}_2$ nanocubes and (d) $\text{Fe}_3\text{O}_4@ \chi\text{-Fe}_5\text{C}_2$ octahedra after 100 h on stream derived from the nitrogen adsorption-desorption isotherms by the BJH method.



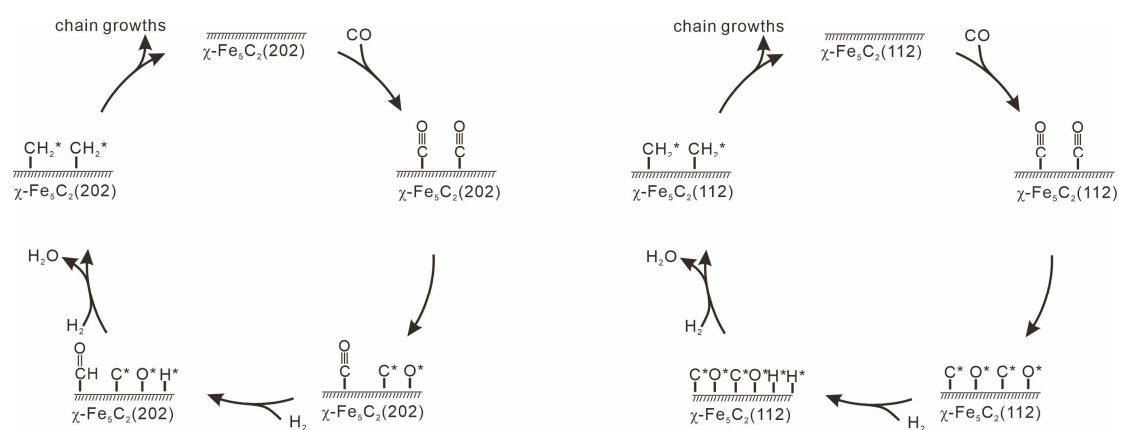
Supplementary Figure 21 | Raman spectra of $\text{Fe}_3\text{O}_4@ \chi\text{-Fe}_5\text{C}_2$ nanocubes (a) and $\text{Fe}_3\text{O}_4@ \chi\text{-Fe}_5\text{C}_2$ octahedra (b) after 100 h on stream. TGA profiles of $\text{Fe}_3\text{O}_4@ \chi\text{-Fe}_5\text{C}_2$ nanocubes (c) and $\text{Fe}_3\text{O}_4@ \chi\text{-Fe}_5\text{C}_2$ octahedra (d) after 100 h on stream.



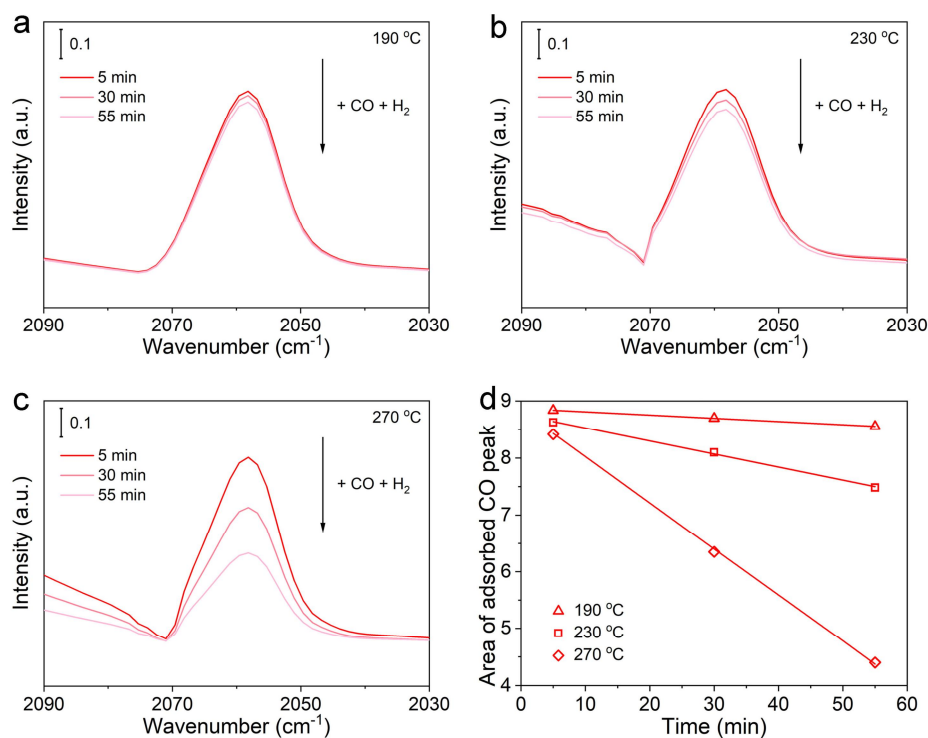
Supplementary Figure 22 | (a) *In-situ* DRIFTS spectra of Fe₃O₄@ χ -Fe₅C₂ nanocubes and (b) Fe₃O₄@ χ -Fe₅C₂ octahedra after being exposed to 20 bar of syngas for 30 min at 270 °C.

Supplementary Table 5 |Assignment of DRIFTS peaks.

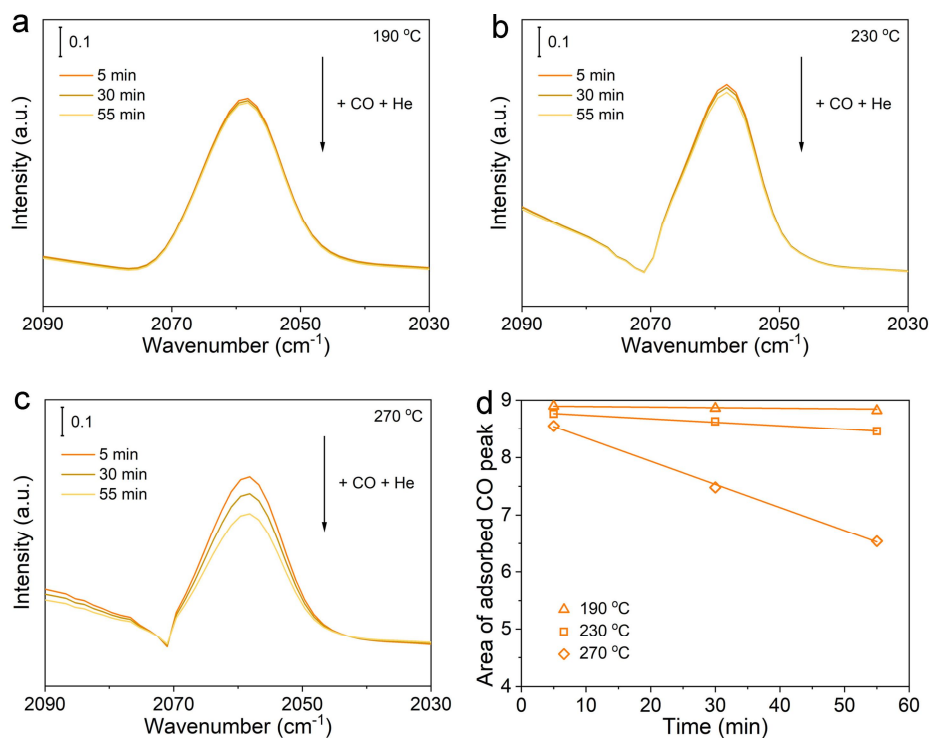
Wavenumber (cm ⁻¹)	Assignment	References
2360, 2336	Gaseous CO ₂	1
2955	Asymmetrical stretching vibration of C-H bonds in CH ₃ *	2, 3
2919	Asymmetrical stretching vibration of C-H bonds in CH ₂ *	4
2849	Symmetrical stretching vibration of C-H bonds in CH ₂ *	5
1741	Stretching vibration of C=O bonds in CHO*	6,7



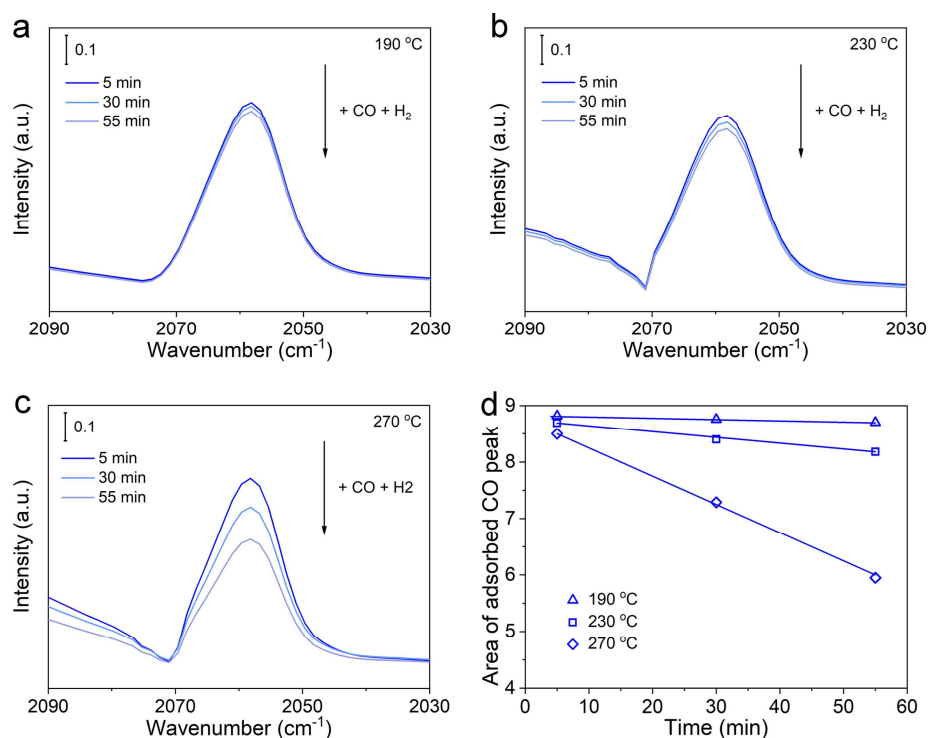
Supplementary Figure 23 | The schematic diagram for different CO dissociation pathways of Fe₃O₄@χ-Fe₅C₂ nanocubes and Fe₃O₄@χ-Fe₅C₂ octahedra.



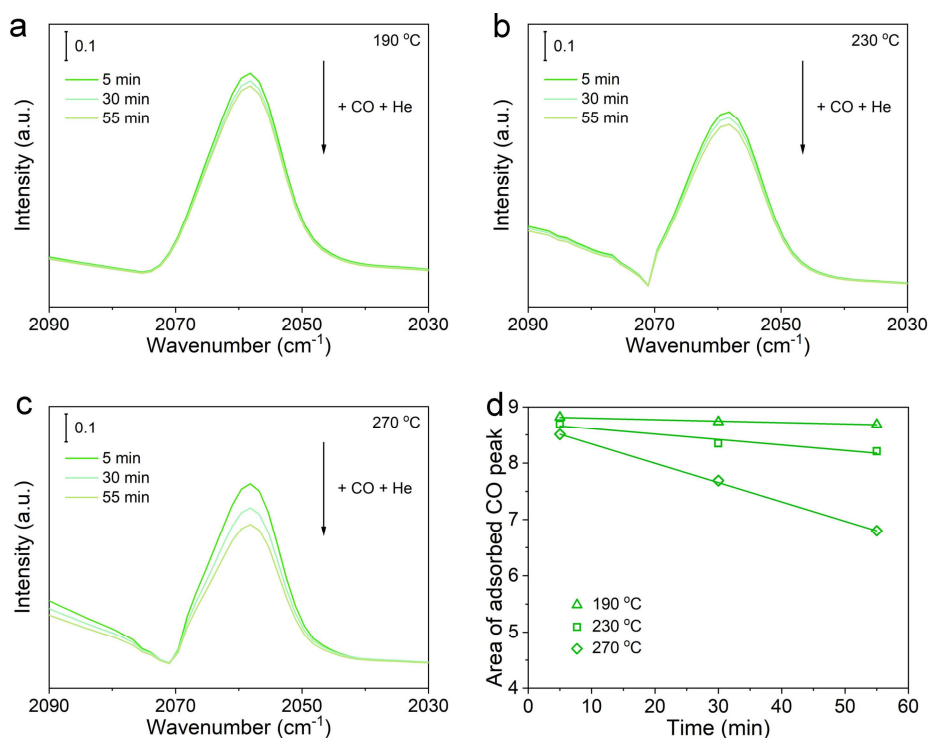
Supplementary Figure 24 | Mechanistic studies of $\text{Fe}_3\text{O}_4@ \chi\text{-Fe}_5\text{C}_2$ nanocubes for CO dissociation. (a-c) *In-situ* DRIFTS spectra of CO adsorption on $\text{Fe}_3\text{O}_4@ \chi\text{-Fe}_5\text{C}_2$ nanocubes after being exposed to CO and purged with H_2 at (a) 190 °C, (b) 230 °C, and (c) 270 °C. Specifically, $\text{Fe}_3\text{O}_4@ \chi\text{-Fe}_5\text{C}_2$ nanocubes were exposed to 1 bar of CO for 30 min. Subsequently, the spectra were recorded at 5, 30, and 55 min after being purged H_2 . (d) Time course of the area of adsorbed CO peak over $\text{Fe}_3\text{O}_4@ \chi\text{-Fe}_5\text{C}_2$ nanocubes in H_2 .



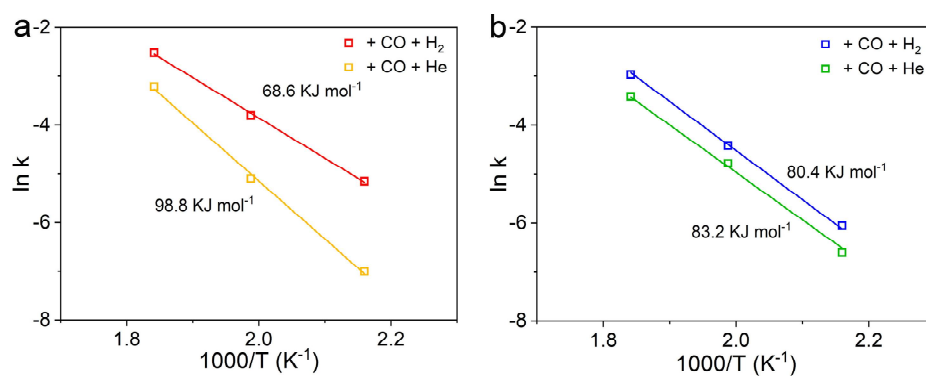
Supplementary Figure 25 | Mechanistic studies of $\text{Fe}_3\text{O}_4@ \chi\text{-Fe}_5\text{C}_2$ nanocubes for CO dissociation. (a-c) *In-situ* DRIFTS spectra of CO adsorption on $\text{Fe}_3\text{O}_4@ \chi\text{-Fe}_5\text{C}_2$ nanocubes after being exposed to CO and purged with He at (a) 190 °C, (b) 230 °C, and (c) 270 °C. Specifically, $\text{Fe}_3\text{O}_4@ \chi\text{-Fe}_5\text{C}_2$ nanocubes were exposed to 1 bar of CO for 30 min. Subsequently, the spectra were recorded at 5, 30, and 55 min after being purged He. (d) Time course of the area of adsorbed CO peak over $\text{Fe}_3\text{O}_4@ \chi\text{-Fe}_5\text{C}_2$ nanocubes in He.



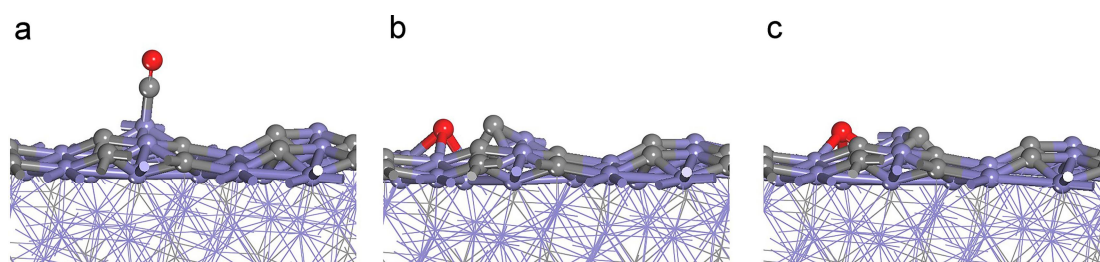
Supplementary Figure 26 | Mechanistic studies of $\text{Fe}_3\text{O}_4@ \chi\text{-Fe}_5\text{C}_2$ octahedra for CO dissociation. (a-c) *In-situ* DRIFTS spectra of CO adsorption on $\text{Fe}_3\text{O}_4@ \chi\text{-Fe}_5\text{C}_2$ octahedra after being exposed to CO and purged with H_2 at (a) 190 °C, (b) 230 °C, and (c) 270 °C. Specifically, $\text{Fe}_3\text{O}_4@ \chi\text{-Fe}_5\text{C}_2$ octahedra were exposed to 1 bar of CO for 30 min. Subsequently, the spectra were recorded at 5, 30, and 55 min after being purged H_2 . (d) Time course of the area of adsorbed CO peak over $\text{Fe}_3\text{O}_4@ \chi\text{-Fe}_5\text{C}_2$ octahedra in H_2 .



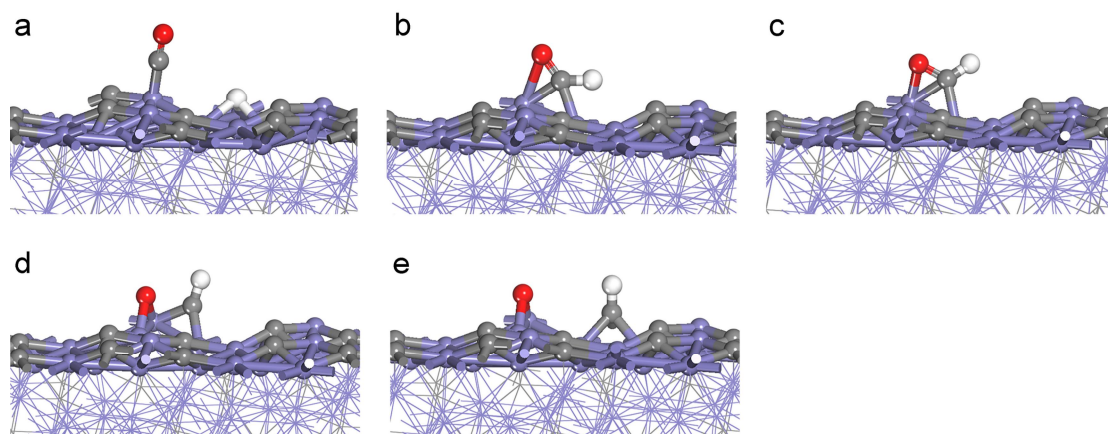
Supplementary Figure 27 | Mechanistic studies of $\text{Fe}_3\text{O}_4@ \chi\text{-Fe}_5\text{C}_2$ octahedra for CO dissociation. (a-c) *In-situ* DRIFTS spectra of CO adsorption on $\text{Fe}_3\text{O}_4@ \chi\text{-Fe}_5\text{C}_2$ octahedra after being exposed to CO and purged with He at (a) 190 °C, (b) 230 °C, and (c) 270 °C. Specifically, $\text{Fe}_3\text{O}_4@ \chi\text{-Fe}_5\text{C}_2$ octahedra were exposed to 1 bar of CO for 30 min. Subsequently, the spectra were recorded at 5, 30, and 55 min after being purged He. (d) Time course of the area of adsorbed CO peak over $\text{Fe}_3\text{O}_4@ \chi\text{-Fe}_5\text{C}_2$ octahedra in He.



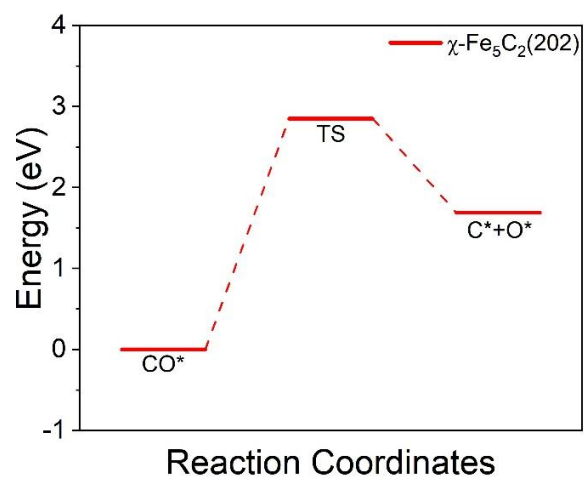
Supplementary Figure 28 | (a, b) Arrhenius plots and activation energy of (a) Fe₃O₄@χ-Fe₅C₂ nanocubes and (b) Fe₃O₄@χ-Fe₅C₂ octahedra after being exposed to CO and purged with He or H₂.



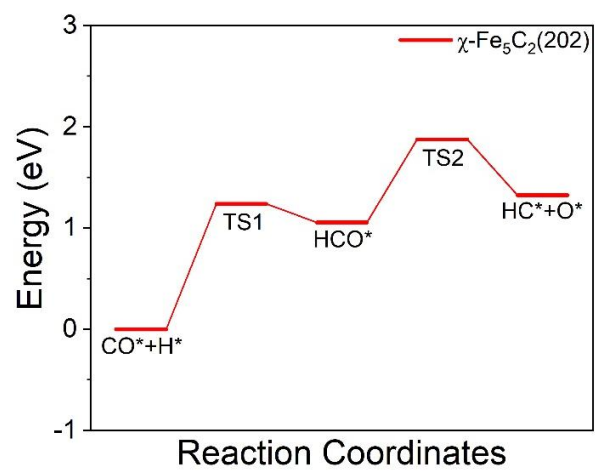
Supplementary Figure 29 | Calculated structure models of the direct CO dissociation route on the χ -Fe₅C₂(202) facet. Models of (a) CO*, (b) TS, and (c) C* + O*.



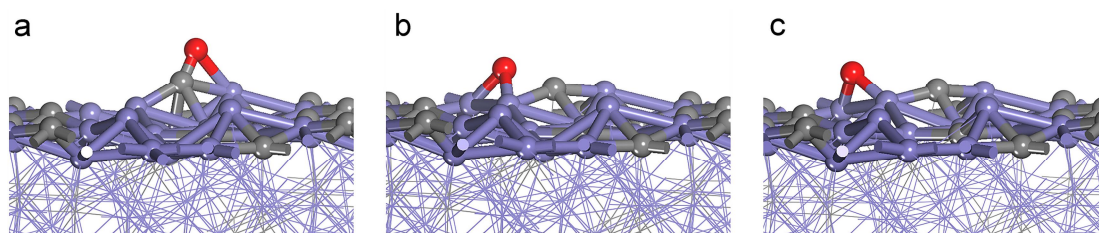
Supplementary Figure 30 | Calculated structure models of the hydrogen-assisted CO dissociation route on the χ -Fe₅C₂(202) facet. Models of (a) CO* + H*, (b) TS1, (c) HCO*, (d) TS2, and (e) HC* + O*.



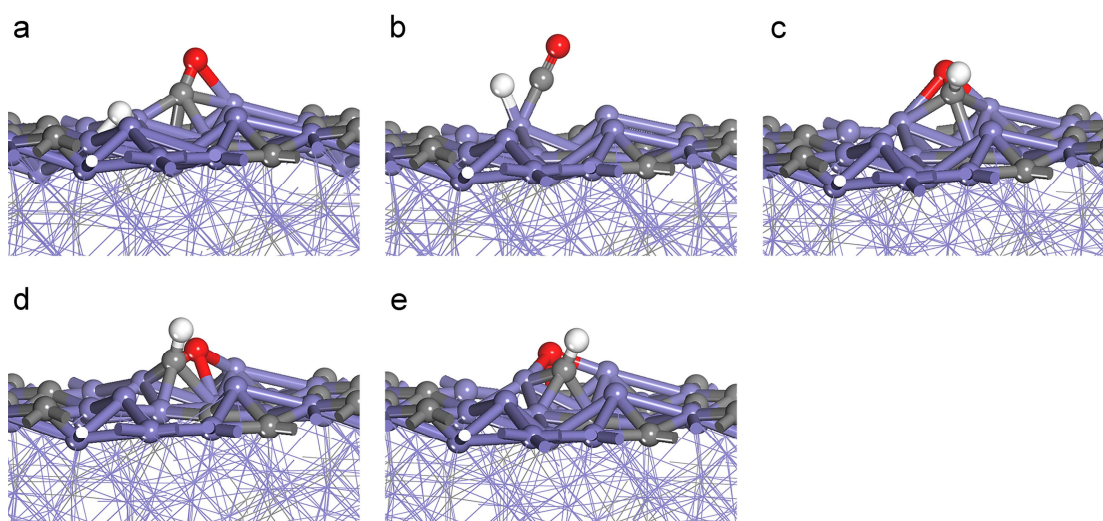
Supplementary Figure 31 | Energy barrier of the direct CO dissociation route on the χ -Fe₅C₂(202) facet.



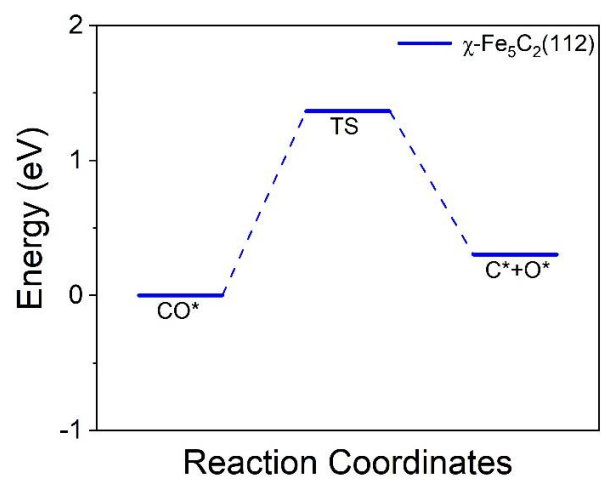
Supplementary Figure 32 | Energy barrier of the hydrogen-assisted CO dissociation route on the $\chi\text{-Fe}_5\text{C}_2(202)$ facet.



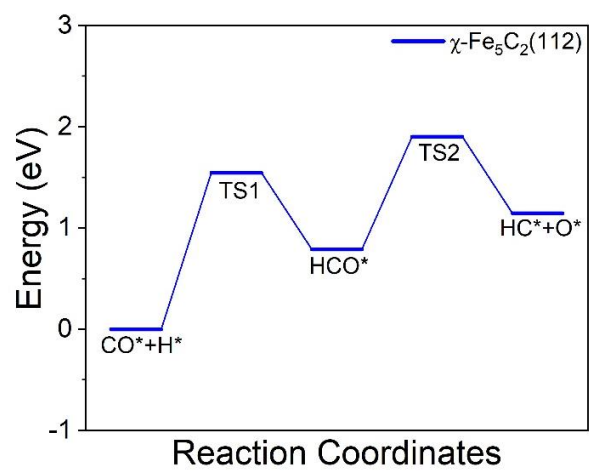
Supplementary Figure 33 | Calculated structure models of the direct CO dissociation route on the χ -Fe₅C₂(112) facet. Models of (a) CO*, (b) TS, and (c) C* + O*.



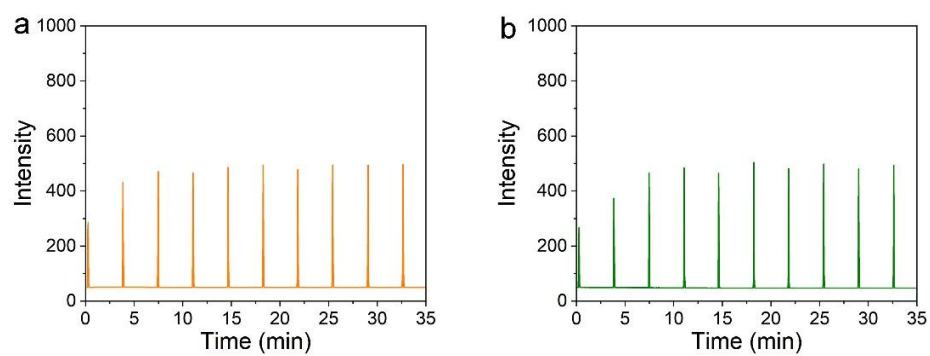
Supplementary Figure 34 | Calculated structure models of the hydrogen-assisted CO dissociation route on the χ -Fe₅C₂(112) facet. Models of (a) CO* + H*, (b) TS1, (c) HCO*, (d) TS2, and (e) HC* + O*.



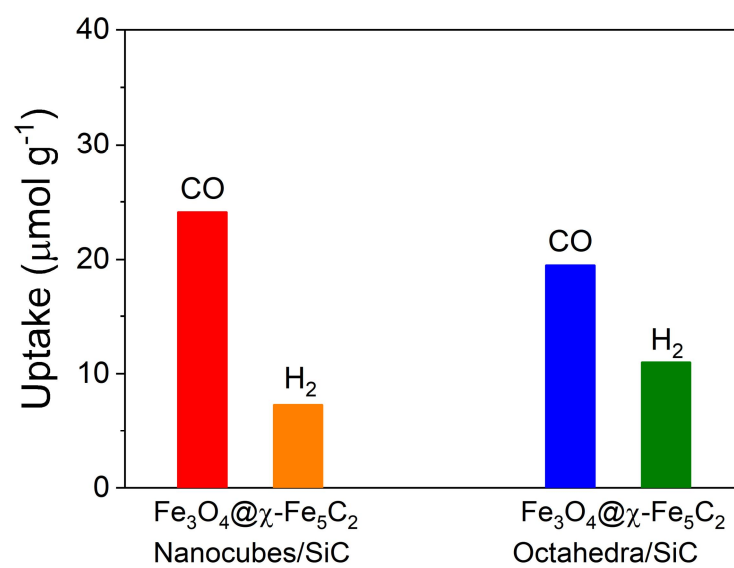
Supplementary Figure 35 | Energy barrier of the direct CO dissociation route on the $\chi\text{-Fe}_5\text{C}_2(112)$ facet.



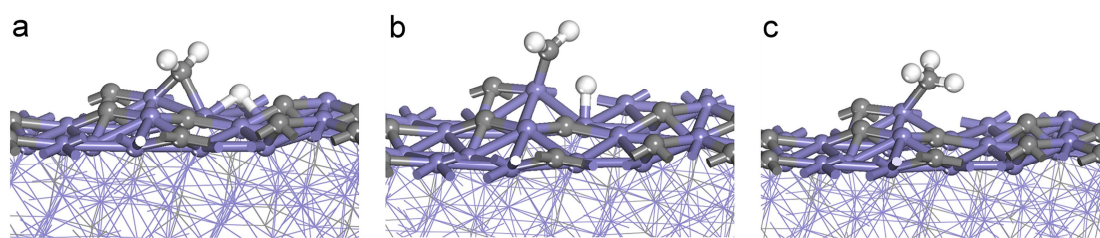
Supplementary Figure 36 | Energy barrier of the hydrogen-assisted CO dissociation route on the $\chi\text{-Fe}_5\text{C}_2(112)$ facet.



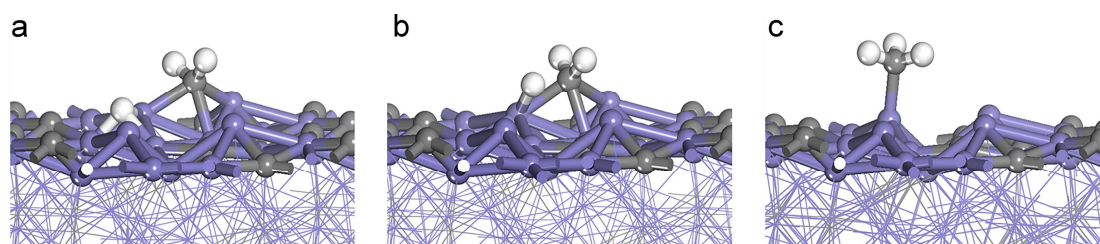
Supplementary Figure 37 | H₂ pulse profiles of (a) Fe₃O₄@ χ -Fe₅C₂ nanocubes/SiC and (b) Fe₃O₄@ χ -Fe₅C₂ octahedra/SiC.



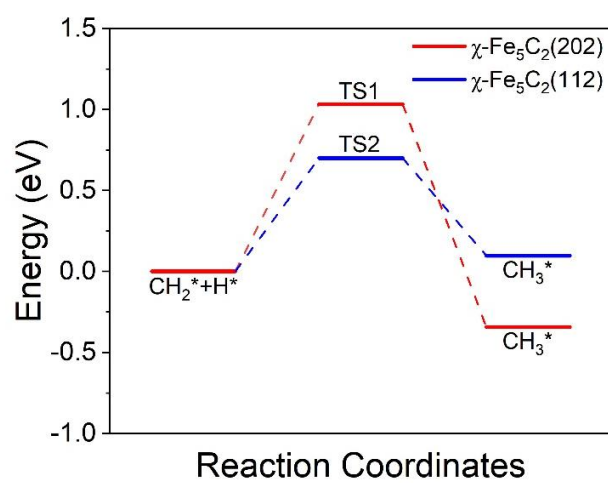
Supplementary Figure 38 | The amount of adsorbed CO and H₂ over Fe₃O₄@ χ -Fe₅C₂ nanocubes/SiC and Fe₃O₄@ χ -Fe₅C₂ octahedra/SiC.



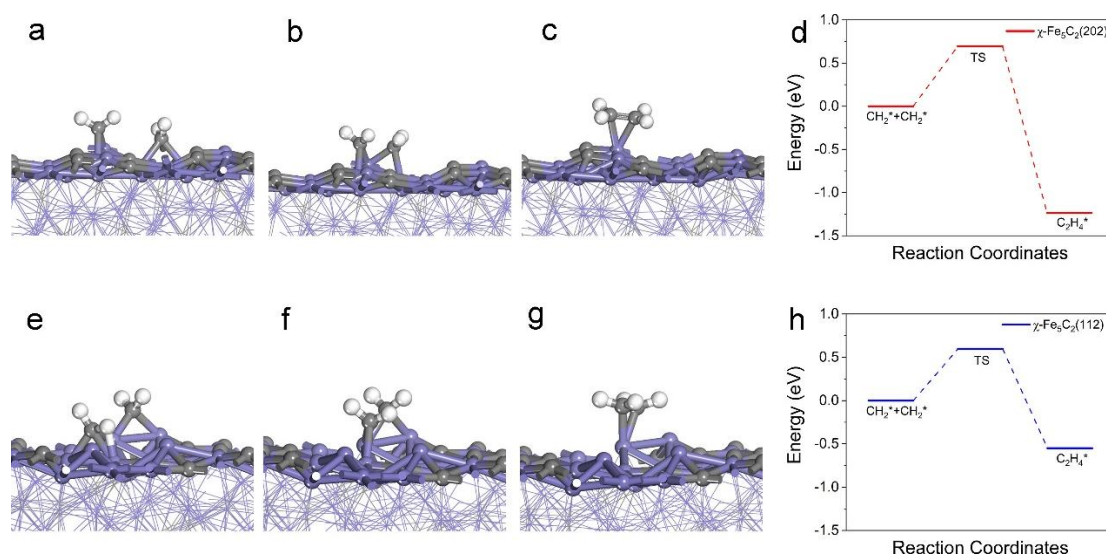
Supplementary Figure 39 | Calculated structure models of CH_2^*+H^* on the $\chi\text{-Fe}_5\text{C}_2(202)$ facet. Models of (a) CH_2^*+H^* , (b) TS1, and (c) CH_3^* .



Supplementary Figure 40 | Calculated structure models of $\text{CH}_2^* + \text{H}^*$ on the $\chi\text{-Fe}_5\text{C}_2(112)$.
Models of (a) $\text{CH}_2^* + \text{H}^*$, (b) TS2, and (c) CH_3^* .



Supplementary Figure 41 | Comparison in energy barriers of CH₂* + H* over χ -Fe₅C₂(202) and χ -Fe₅C₂(112) facets.



Supplementary Figure 42 | Calculated structure models and energy barriers of $\text{CH}_2^* + \text{CH}_2^*$. Models of (a) $\text{CH}_2^* + \text{CH}_2^*$, (b) TS, (c) C_2H_4^* over $\chi\text{-Fe}_5\text{C}_2(202)$. (d) Energy barriers of $\text{CH}_2^* + \text{CH}_2^*$ over $\chi\text{-Fe}_5\text{C}_2(202)$. Models of (a) $\text{CH}_2^* + \text{CH}_2^*$, (b) TS, (c) C_2H_4^* over $\chi\text{-Fe}_5\text{C}_2(112)$. (d) Energy barriers of $\text{CH}_2^* + \text{CH}_2^*$ over $\chi\text{-Fe}_5\text{C}_2(112)$.

Supplementary References

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